

PLAYFAIR CIPHER

1. INTRODUCTION:

Playfair Cipher is a classical encryption technique that encrypts pairs of letters (digraphs) instead of single letters. It was invented in 1854 by Charles Wheatstone and promoted by Lord Playfair. It is more secure than simple substitution ciphers because it hides letter frequency patterns.

2. CONCEPT:

- 1 Encrypts letters in pairs (digraphs)
- 2 Uses a 5×5 matrix built from a keyword
- 3 Combines I and J into one cell
- 4 Applies rules based on position of letters in matrix
- 5 Used for manual encryption before computers

3. KEY MATRIX EXAMPLE:

Keyword: MONARCHY

M O N A R
C H Y B D
E F G I K
L P Q S T
U V W X Z

4. IMPLEMENTATION:

- 1 Choose a keyword
- 2 Create 5×5 matrix (remove duplicates, combine I/J)
- 3 Split plaintext into pairs
- 4 If letters repeat, insert 'X'
- 5 Apply rules (same row, same column, rectangle rule)
- 6 Combine results to form ciphertext

5. EXAMPLE:

Plaintext: HELLO

Pairs: HE LX LO

After encryption → Ciphertext: (varies based on matrix)

6. REAL TIME APPLICATIONS:

- 1 Used in military communication (historical)
- 2 Educational purpose to understand encryption
- 3 Basic cryptography learning
- 4 Puzzle and coding competitions